

The resting membrane potential

The written notes for the resting membrane potential walkthrough. Week 4 competencies 1 to 5.

To print: these notes as a two-column PDF. Already printed the Week 4 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

How much of each ion

Ion	Inside (mM)	Outside (mM)
K ⁺	150	5
Na ⁺	15	145
Cl ⁻	10	108
Ca ²⁺	0.0001	1
Large anions (proteins, ATP)	Many	Almost none

- Values vary slightly between textbooks, but the pattern is the same.
- Potassium is the only ion in the table that is more concentrated inside than outside. Sodium, chloride and calcium are all more concentrated outside.
- The large anions are too big to cross the membrane. They are the negative charge left behind when potassium leaves.
- Ions cannot cross the lipid bilayer on their own. They need channels, which are proteins built from the instructions in genes.

How the inside becomes negative

1. The membrane has many potassium leak channels, which are always open.
2. Potassium is more concentrated inside, so it diffuses out through them, down its concentration gradient.
3. Each potassium ion carries a positive charge out and leaves an unpaired negative charge behind, so the inside becomes negative. The separated charge collects in a very thin layer along the membrane, and only a tiny fraction of the cell's potassium has to leave to produce it.
4. The negative inside attracts potassium back in, so the electrical gradient on potassium acts inward. It grows as more potassium leaves.
5. When the electrical gradient is equal and opposite to the concentration gradient, potassium moves in and out at equal rates, with no net movement. That membrane potential is potassium's equilibrium potential, about -90 mV.

6. A few open sodium channels let a small amount of sodium in, carrying positive charge in and making the inside slightly less negative. The resting potential is about -70 mV.

- Permeability is how easily an ion crosses the membrane, and it depends mainly on how many of that ion's channels are open. The ion with the highest permeability has the most influence on the membrane potential. At rest a neuron is roughly 25 times more permeable to potassium than to sodium.
- If every potassium channel were blocked, potassium could no longer carry positive charge out, but sodium would keep leaking in, so the inside would become less negative. The potassium leak channels are the main reason the inside is negative at all.

Resting membrane potential

The voltage across the membrane of a cell at rest, inside relative to outside, about -70 mV in a neuron.

Concentration gradient

The difference in an ion's concentration across the membrane, also called the chemical gradient. It depends only on concentration, not on charge.

Electrical gradient

The effect of the membrane potential on a charged ion. It depends only on the ion's charge and the membrane potential, not on concentration.

Electrochemical gradient

The concentration gradient and the electrical gradient together. It determines the direction of an ion's net movement.

Equilibrium potential

The membrane potential at which the electrical gradient on an ion is equal and opposite to its concentration gradient, so the ion has no net movement.

Permeability

How easily an ion crosses the membrane, set mainly by how many of its channels are open.

The electrochemical gradient

Ion	Concentration gradient	Electrical gradient at -70 mV	Result
K ⁺	Outward	Inward	They oppose each other; a small net outward force
Na ⁺	Inward	Inward	Both act inward; a large net inward force

Ion	Equilibrium potential (about)
K ⁺	-90 mV
Na ⁺	+60 mV
Cl ⁻	-65 mV
Ca ²⁺	+120 mV

- The difference between the membrane potential and an ion's equilibrium potential is the driving force on that ion.
- At -70 mV, potassium is 20 mV from its equilibrium potential and sodium is 130 mV from its, so the driving force on sodium is much larger.
- The membrane potential is measured with a microelectrode inside the cell and a reference electrode outside, and it is always stated as the inside relative to the outside.

Driving force

The difference between the membrane potential and an ion's equilibrium potential.

Which way will it move?

- When an ion's channels open, net movement of that ion drives the membrane potential toward that ion's equilibrium potential.
- When its channels are closed, the ion cannot cross, however large its electrochemical gradient.
- At -70 mV, if sodium channels open, sodium moves in, because its concentration gradient and the electrical gradient both act inward. The membrane depolarizes toward +60 mV.
- At +30 mV, if potassium channels open, potassium moves out with a large driving force: 120 mV from its equilibrium potential, compared with 20 mV at rest. Potassium carries positive charge out and the membrane potential falls. This is how the membrane repolarizes during an action potential.

The Nernst and Goldman equations

- In this course you read and interpret these equations; you do not calculate with them.
- The Nernst equation calculates one ion's equilibrium potential from its concentrations outside and inside the cell. Potassium at 150 mM inside and 5 mM outside gives about -90 mV.
- If outside potassium rises from 5 to 10 mM, potassium's equilibrium potential becomes less negative, about -72 mV, because the concentration gradient driving potassium out is smaller. This is why high blood potassium depolarizes cells.
- The Goldman equation calculates the membrane potential from the concentrations of several ions and the membrane's permeability to each. It works much like a weighted average of the equilibrium potentials, with each ion weighted by its permeability.
- Potassium's permeability is by far the largest, so the resting potential lies near -90 mV, and sodium's small permeability shifts it to about -70 mV.
- Opening more potassium leak channels moves the membrane potential closer to -90 mV (hyperpolarization). Opening many sodium channels makes sodium permeability exceed potassium permeability, so the membrane potential rises toward +60 mV. This rapid depolarization is the rising phase of an action potential.

What the pump does and does not do

- The sodium-potassium pump (Na⁺/K⁺ ATPase) uses energy from ATP to move ions against their concentration gradients: for each ATP it hydrolyzes, it moves 3 Na⁺ out and 2 K⁺ in.
- Its main job is to maintain the concentration gradients over minutes to hours, because sodium leaks in and potassium leaks out continuously.
- Because each cycle moves 3 positive charges out and only 2 in, the pump also makes the inside slightly more negative directly, but only by a few millivolts.
- It does not set the -70 mV resting potential from moment to moment; the leak channels do that. If the pump stops, the membrane potential changes by only a few millivolts at first. Over hours it drifts toward 0 mV as the concentration gradients run down.

Terms for voltage changes

Depolarization

The membrane potential becomes less negative than rest (moves toward zero or above).

Overshoot

The part of an action potential where the inside is positive.

Repolarization

The membrane potential returns toward its resting value, even while the inside is still positive.

Hyperpolarization

The membrane potential becomes more negative than rest.

- These terms describe the direction of the change relative to the resting potential, not whether the inside is positive or negative.
- Channels come in four kinds: leak channels (no gate, always open), voltage-gated, ligand-gated (chemically gated) and mechanically gated. The channel gating walkthrough covers these in detail.

Resting potential and excitability

- A neuron fires an action potential when its membrane is depolarized to threshold, about -55 mV. The difference between the resting potential and threshold is the depolarization needed to fire, and it determines how excitable the cell is.
- A cell that is depolarized is usually more excitable. If a cell stays depolarized for a long time, the inactivation gates of its voltage-gated sodium channels stay closed, fewer channels are available to open, and the cell becomes less excitable.

Condition	Blood K ⁺ (normal 3.5 to 5.0 mM)	K ⁺ equilibrium potential	Resting potential	Effect
Hyperkalemia (missed dialysis)	7.0 mM	About -81 mV	About -65 mV, depolarized	At first closer to threshold; if it lasts, sodium channels stay inactivated and cells can barely fire action potentials. In the heart: slow, disorganized conduction and risk of cardiac arrest.
Hypokalemia (loop diuretic)	2.8 mM	About -105 mV	About -76 mV, hyperpolarized	Farther from threshold, so a larger depolarization is needed to fire: muscle weakness and cramps. Low potassium also makes digoxin toxicity more likely.

- Treating hyperkalemia: IV calcium gluconate protects the heart by raising the threshold of cardiac cells, without changing the potassium level. Insulin with glucose lowers blood potassium by stimulating the sodium-potassium pump, which moves potassium into cells; the potassium is still in the body, so the effect lasts only a few hours. Dialysis is the only one of the three that removes potassium from the body.
- Lidocaine blocks voltage-gated sodium channels, not the leak channels, so the resting potential barely changes. A stimulus still depolarizes the membrane, but sodium cannot pass through the blocked channels, so no action potentials are generated and no pain is felt. A normal resting potential is necessary for a neuron to signal, but it is not sufficient on its own.